Experiment:1

AIM: Build a responsive web application for shopping cart with registration, login, catalog and cart pages using CSS3 features, flex and grid.

SOLUTION:

Responsive design means building a website, it automatically adjusts to different screen sizes and devices - like mobile phones, tablets, laptops, and desktops.

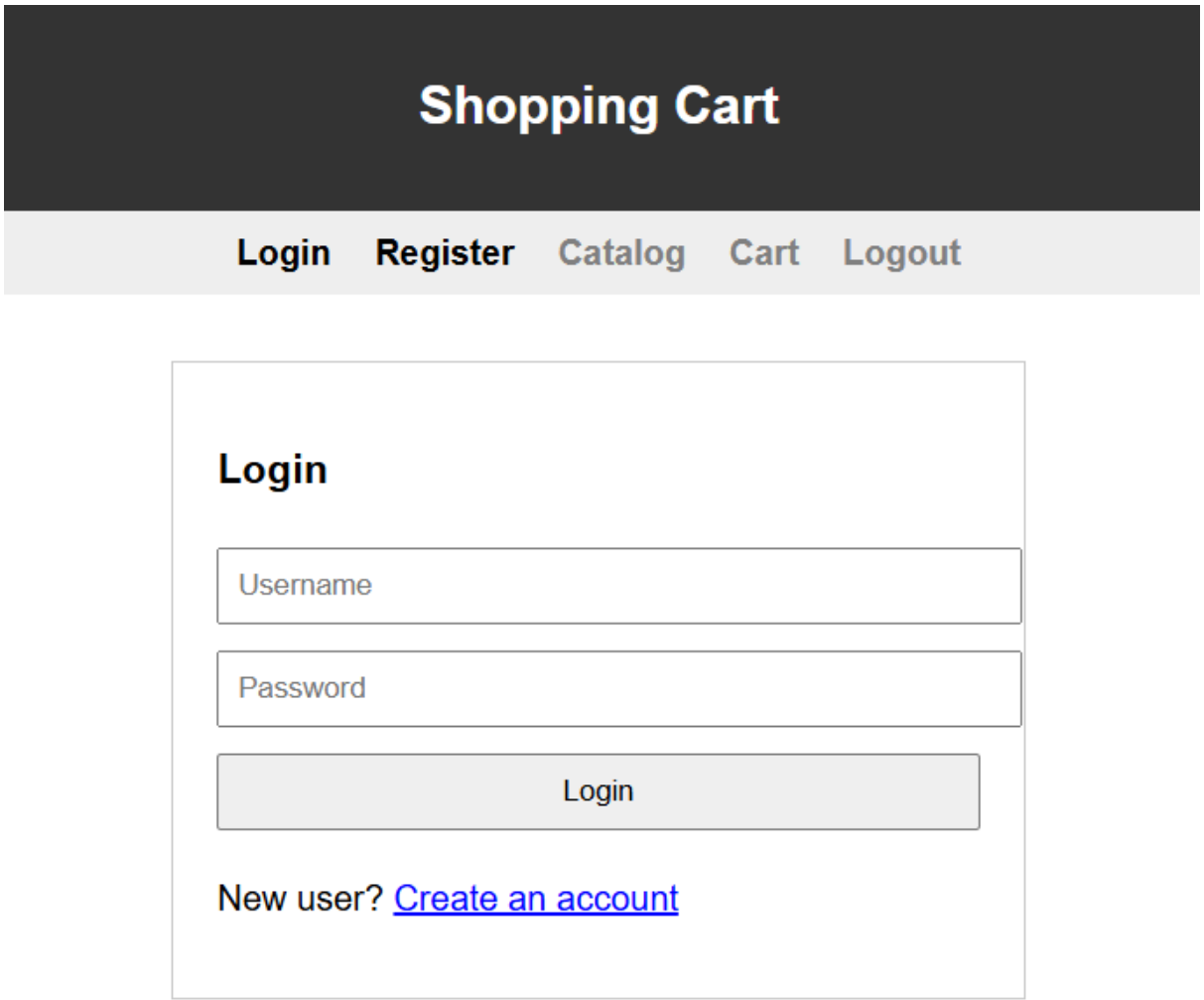
Source Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Login</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body onload="updateNav()">
<header><h2>Shopping Cart</h2></header>
<nav>
<a id="navLogin" href="index.html">Login</a>
<a id="navRegister" href="register.html">Register</a>
<a id="navCatalog" class="disabled" href="catalog.html">Catalog</a>
<a id="navCart" class="disabled" href="cart.html">Cart</a>
<a id="navLogout" class="disabled" href="logout.html">Logout</a>
</nav>
<div class="form-box">
<h3>Login</h3>
<input id="logUser" placeholder="Username">
<input id="logPass" type="password" placeholder="Password">
<button onclick="login()">Login</button>
<p id="logMsg"></p>
<p>New user? <span class="link"
onclick="location.href='register.html'">Create an account</span></p>
</div>
<script src="js/app.js"></script>
```

```
</body>
</html>
```

Output:

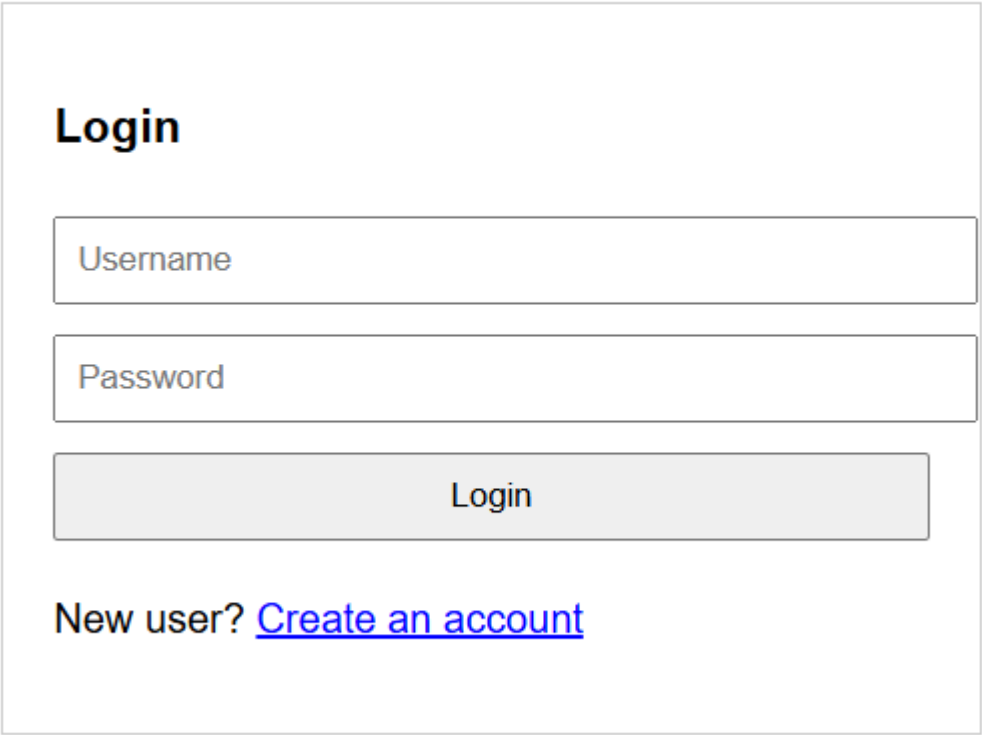


Login.html

```
<!DOCTYPE html>
<html>
<head>
<title>Login</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body onload="navSetup()">
<header><h2>Shopping Cart</h2></header>
<nav>
<a id="navLogin" href="login.html">Login</a>
<a id="navRegister" href="register.html">Register</a>
```

```
<a id="navCatalog" class="disabled" href="catalog.html">Catalog</a>
<a id="navCart" class="disabled" href="cart.html">Cart</a>
<a id="navLogout" class="disabled" href="logout.html">Logout</a>
</nav>
<div class="form-box">
<h3>Login</h3>
<input id="logUser" placeholder="Username">
<input id="logPass" type="password" placeholder="Password">
<button onclick="login()">Login</button>
<p id="logMsg"></p>
<p>New user? <span class="link" onclick="location.href='register.html'">Create an account</span></p>
</div>
<script src="js/app.js"></script>
</body>
</html>
```

Output:

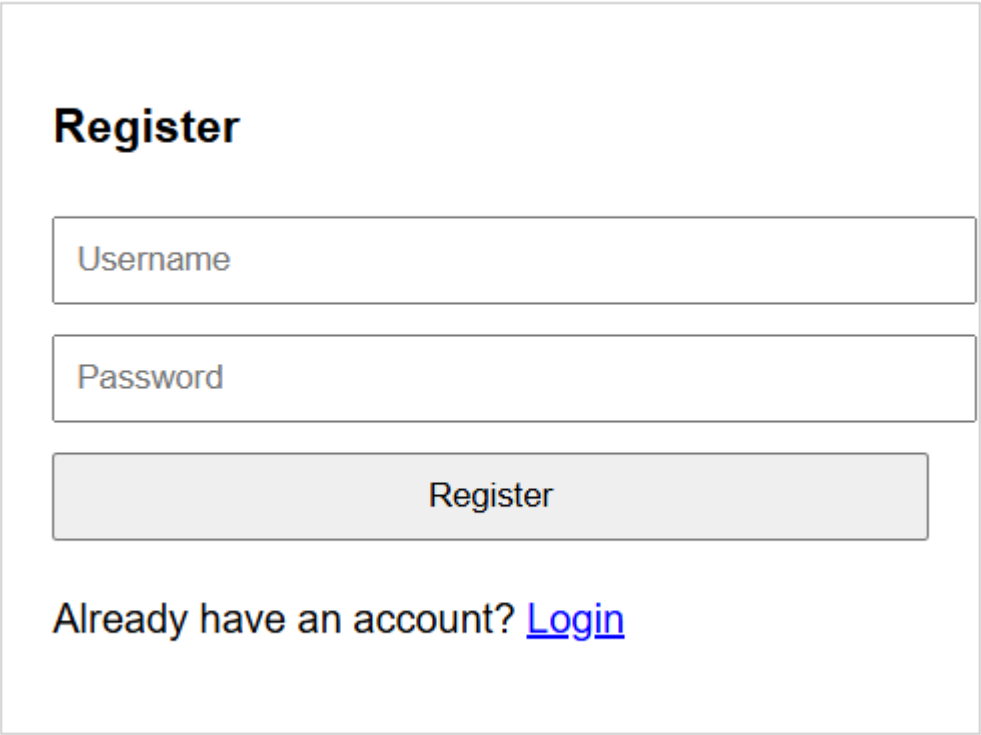


Register.html

```
<!DOCTYPE html>
```

```html
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Register</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body onload="updateNav()">
<header><h2>Shopping Cart</h2></header>
<nav>
<a id="navLogin" href="index.html">Login</a>
<a id="navRegister" href="register.html">Register</a>
<a id="navCatalog" class="disabled" href="catalog.html">Catalog</a>
<a id="navCart" class="disabled" href="cart.html">Cart</a>
<a id="navLogout" class="disabled" href="logout.html">Logout</a>
</nav>
<div class="form-box">
<h3>Register</h3>
<input id="regUser" placeholder="Username">
<input id="regPass" type="password" placeholder="Password">
<button onclick="register()">Register</button>
<p id="regMsg"></p>
<p>Already have an account?
<span class="link" onclick="location.href='index.html'">Login</span>
</p>
</div>
<script src="js/app.js"></script>
</body>
</html>
```

Output:

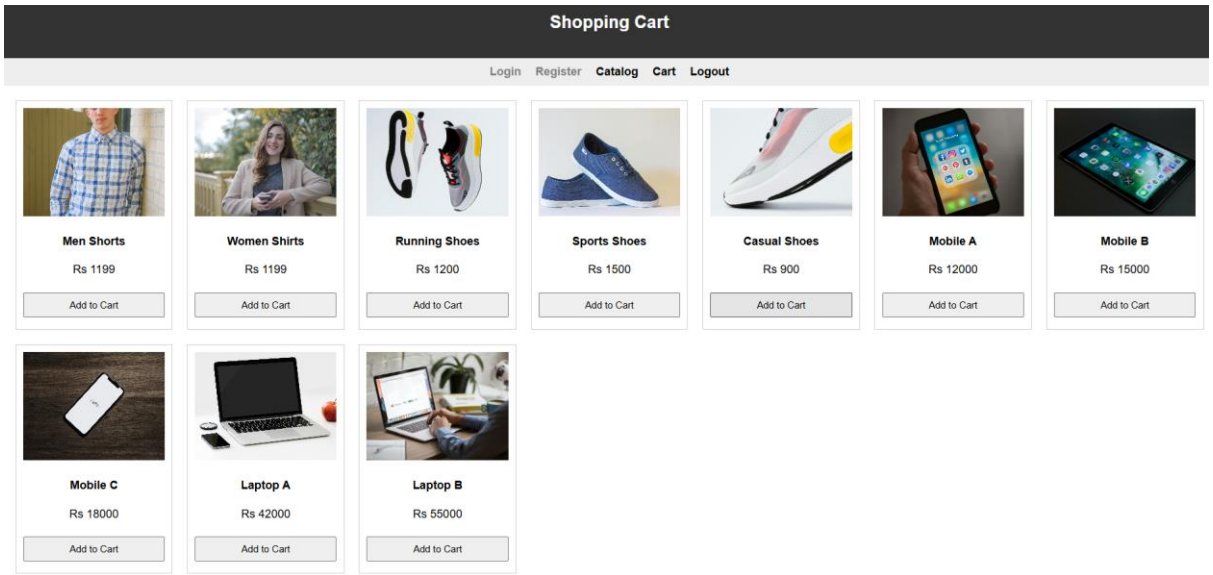


Catalog.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Catalog</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body onload="updateNav(); loadProducts();">
<header><h2>Shopping Cart</h2></header>
<nav>
<a id="navLogin" href="index.html">Login</a>
<a id="navRegister" href="register.html">Register</a>
<a id="navCatalog" class="disabled" href="catalog.html">Catalog</a>
<a id="navCart" href="cart.html">Cart</a>
<a id="navLogout" href="logout.html">Logout</a>
</nav>
<div class="grid" id="productGrid"></div>
```

```
<script src="js/app.js"></script>
</body>
</html>
```

Output:

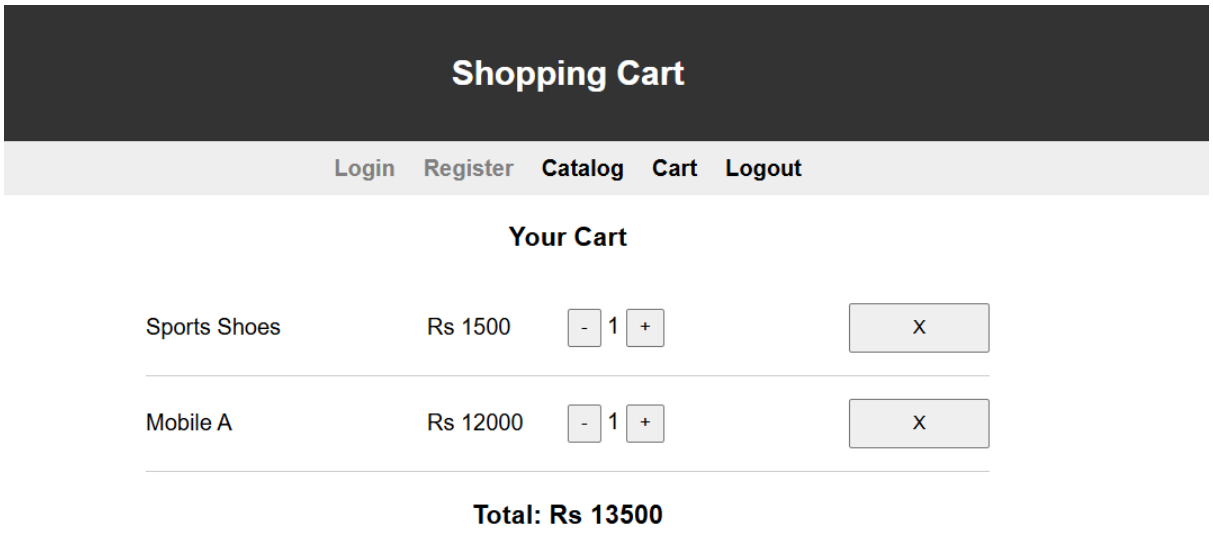


Cart.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Cart</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body onload="updateNav(); loadCart();">
<header><h2>Shopping Cart</h2></header>
<nav>
<a id="navLogin" href="index.html">Login</a>
<a id="navRegister" href="register.html">Register</a>
<a id="navCatalog" href="catalog.html">Catalog</a>
<a id="navCart" class="disabled" href="cart.html">Cart</a>
<a id="navLogout" href="logout.html">Logout</a>
</nav>
<h3 style="text-align:center">Your Cart</h3>
<div class="cart-box" id="cartList"></div>
```

```
<h3 style="text-align:center">Total: Rs <span id="totalAmount">0</span></h3>
<script src="js/app.js"></script>
</body>
</html>
```

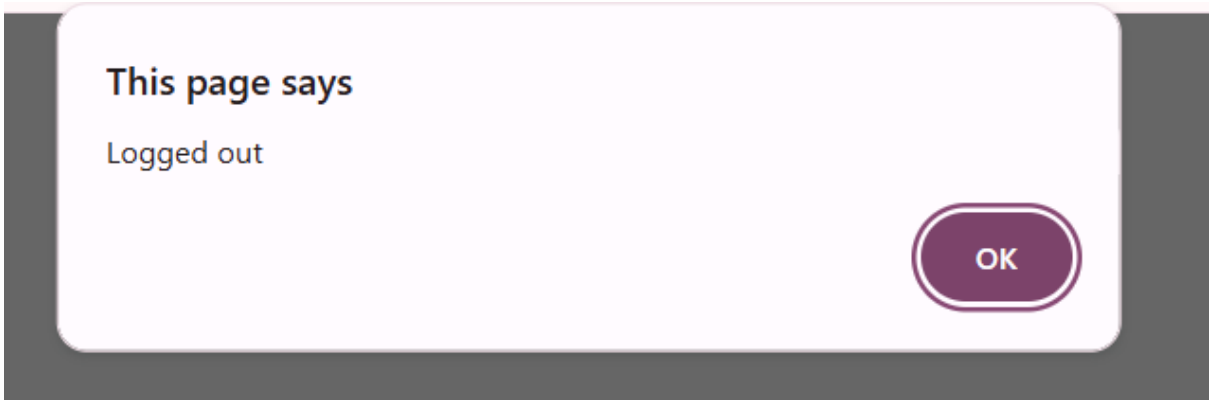
Output:



Logout.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Logout</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body onload="logout()"></body>
<script src="js/app.js"></script>
</html>
```

Output:



Js/app.js

```
let isLoggedIn = localStorage.getItem("isLoggedIn") === "true";
let cart = JSON.parse(localStorage.getItem("cart")) || [];

// ---------- NAV CONTROL ----------
function updateNav() {
  if (isLoggedIn) {
    navRegister.classList.add("disabled");
    navLogin.classList.add("disabled");
    navCatalog.classList.remove("disabled");
    navCart.classList.remove("disabled");
    navLogout.classList.remove("disabled");
  } else {
    navRegister.classList.remove("disabled");
    navLogin.classList.remove("disabled");
    navCatalog.classList.add("disabled");
    navCart.classList.add("disabled");
    navLogout.classList.add("disabled");
  }
}

// ---------- REGISTER ----------
function register() {
  let username = regUser.value.trim();
  let password = regPass.value.trim();
  localStorage.setItem("user", username);
  localStorage.setItem("pass", password);
```

```
  regMsg.innerText = "Registered successfully. Please login.";
  setTimeout(()=>location.href="index.html",800);
}

// ---------- LOGIN ----------
function login() {
  if (
    logUser.value === localStorage.getItem("user") &&
    logPass.value === localStorage.getItem("pass")
  ) {
    isLoggedIn = true;
    localStorage.setItem("isLoggedIn","true");
    location.href="catalog.html";
  } else {
    logMsg.innerText = "Invalid credentials";
  }
}

// ---------- LOGOUT ----------
function logout() {
  localStorage.removeItem("isLoggedIn");
  localStorage.removeItem("cart");
  alert("Logged out");
  location.href="index.html";
}

// ---------- PRODUCTS ----------
let products = [
 {id:1,              name:'Men              Shorts',              price:1199,
img:'https://images.pexels.com/photos/769749/pexels-photo-769749.jpeg'},
 {id:2,             name:'Women             Shirts',             price:1199,
img:'https://images.pexels.com/photos/3775087/pexels-photo-3775087.jpeg'},
 {id:3,             name:'Running             Shoes',             price:1200,
img:'https://images.pexels.com/photos/2529148/pexels-photo-2529148.jpeg'},
```

```
 {id:4,             name:'Sports             Shoes',             price:1500,
img:'https://images.pexels.com/photos/19090/pexels-photo.jpg'},
 {id:5,             name:'Casual             Shoes',             price:900,
img:'https://images.pexels.com/photos/2529147/pexels-photo-2529147.jpeg'},
 {id:6,              name:'Mobile              A',              price:12000,
img:'https://images.pexels.com/photos/607812/pexels-photo-607812.jpeg'},
 {id:7,              name:'Mobile              B',              price:15000,
img:'https://images.pexels.com/photos/1334597/pexels-photo-1334597.jpeg'},
 {id:8,              name:'Mobile              C',              price:18000,
img:'https://images.pexels.com/photos/699122/pexels-photo-699122.jpeg'},
 {id:9,              name:'Laptop              A',              price:42000,
img:'https://images.pexels.com/photos/18105/pexels-photo.jpg'},
 {id:10,              name:'Laptop              B',              price:55000,
img:'https://images.pexels.com/photos/374074/pexels-photo-374074.jpeg'}
];

function loadProducts(){
  products.forEach(p => {
    productGrid.innerHTML += `
      <div class="item">
        <img src="${p.img}">
        <h4>${p.name}</h4>
        <p>Rs ${p.price}</p>
        <button onclick="addToCart(${p.id})">Add to Cart</button>
      </div>`;
  });
}

// ---------- CART ----------
function addToCart(id) {
  let item = cart.find(c => c.id === id);
  if (item) item.qty++;
  else {
    let p = products.find(p => p.id === id);
    cart.push({id:p.id,name:p.name,price:p.price,qty:1});
    alert(p.name+" added to cart");
```

```
  }
  localStorage.setItem("cart",JSON.stringify(cart));
}

function loadCart() {
  cartList.innerHTML="";
  let total=0;
  cart.forEach((c,i)=>{
    cartList.innerHTML+=`
      <div class="cart-item">
        <span>${c.name}</span>
        <span>Rs ${c.price}</span>
        <div>
          <button class="qty-btn" onclick="changeQty(${i},-1)">-</button>
          ${c.qty}
          <button class="qty-btn" onclick="changeQty(${i},1)">+</button>
        </div>
        <button onclick="removeItem(${i})">X</button>
      </div>`;
    total+=c.price*c.qty;
  });
  totalAmount.innerText=total;
}

function changeQty(i,val){
  cart[i].qty+=val;
  if(cart[i].qty<=0) cart.splice(i,1);
  localStorage.setItem("cart",JSON.stringify(cart));
  loadCart();
}

function removeItem(i){
  cart.splice(i,1);
  localStorage.setItem("cart",JSON.stringify(cart));
  loadCart();
}
```

Css/style.css

```
body { font-family: Arial; margin:0; }
header { background:#333; color:#fff; padding:15px; text-align:center; }

nav {
  display:flex;
  justify-content:center;
  gap:20px;
  background:#eee;
  padding:10px;
}
nav a {
  font-weight:bold;
  cursor:pointer;
  text-decoration:none;
  color:black;
}

nav a.disabled {
  color:gray;
  pointer-events:none;
}

.form-box {
  max-width:350px;
  margin:30px auto;
  border:1px solid #ccc;
  padding:20px;
}

input, button {
  width:100%;
  padding:8px;
  margin:6px 0;
}
```

```
.link {
  color:blue;
  text-decoration:underline;
  cursor:pointer;
}

.grid {
  display:grid;
  grid-template-columns:repeat(auto-fit,minmax(200px,1fr));
  gap:20px;
  padding:20px;
}

.item {
  border:1px solid #ccc;
  padding:10px;
  text-align:center;
}

.item img {
  width:100%;
  height:150px;
  object-fit:cover;
}

.cart-box {
  max-width:600px;
  margin:20px auto;
}

.cart-item {
  display:grid;
  grid-template-columns: 2fr 1fr 2fr 1fr;
  align-items:center;
  border-bottom:1px solid #ccc;
  padding:10px 0;
```

```
}

.qty-btn {
  width:auto;
  padding:4px 8px;
}

/* --------- MEDIA QUERY FOR NAV --------- */
@media(max-width:768px){
  nav {
    flex-direction:column;
  }
}
```

Experiment 2:

AIM: Build a **responsive web application** for shopping cart with registration, login, catalog and cart pages **using Bootstrap framework**

SOLUTION:

Bootstrap is a front-end framework that provides ready-made CSS styles and JavaScript components, which we use inside HTML pages.

Make websites responsive:

Bootstrap automatically adjusts layout for:

mobile

tablet

laptop

desktop

Source Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Shopping Cart</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.
css" rel="stylesheet">
<link rel="stylesheet" href="css/style.css">
</head>
<body>

<!-- NAVBAR -->
<nav class="navbar navbar-dark bg-primary navbar-expand-md px-3">
  <div class="container ">
  <a class="navbar-brand" href="#" id="brand">Shopping Cart</a>

  <div class="d-flex flex-column flex-md-row gap-2">
    <button      id="navLogin"      class="btn      btn-light      btn-sm"
onclick="showPage('login')">Login</button>
    <button     id="navRegister"     class="btn     btn-light     btn-sm"
onclick="showPage('register')">Register</button>
    <button             class="btn             btn-light             btn-sm"
onclick="showPage('catalog')">Catalog</button>
    <button class="btn btn-light btn-sm" onclick="showPage('cart')">
      Cart (<span id="cartCount">0</span>)
    </button>
    <button   id="navLogout"   class="btn   btn-danger   btn-sm   d-none"
onclick="logout()">Logout</button>
```

```
  </div>
</div>
</nav>

<div class="container mt-4">

  <!-- LOGIN PAGE -->
  <div id="login" class="page">
    <div class="card p-4 mx-auto" style="max-width:400px">
      <h4 class="mb-3">Login</h4>
      <input        id="loginUser"        class="form-control        mb-2"
placeholder="Username">
      <input  id="loginPass"  type="password"  class="form-control  mb-2"
placeholder="Password">
      <button           class="btn           btn-primary           w-100"
onclick="login()">Login</button>
      <p class="mt-2" id="loginMsg"></p>
      <p class="mt-2 text-center">
        New  user?  <a  href="#"  onclick="showPage('register')">Create  an
account</a>
      </p>
    </div>
  </div>

  <!-- REGISTER PAGE -->
  <div id="register" class="page">
    <div class="card p-4 mx-auto" style="max-width:400px">
      <h4 class="mb-3">Register</h4>
      <input id="regUser" class="form-control mb-2" placeholder="Username">
      <input   id="regPass"   type="password"   class="form-control   mb-2"
placeholder="Password">
      <button           class="btn           btn-primary           w-100"
onclick="register()">Register</button>
      <p class="mt-2" id="regMsg"></p>
      <p class="mt-2 text-center">
        Already       have       an       account?       <a       href="#"
onclick="showPage('login')">Login</a>
      </p>
    </div>
  </div>

  <!-- CATALOG -->
  <div id="catalog" class="page">
    <div class="row" id="productList"></div>
  </div>

  <!-- CART -->
  <div id="cart" class="page">
    <h4>Your Cart</h4>
    <div id="cartItems"></div>
    <h5 class="mt-3">Total: ₹ <span id="total">0</span></h5>
  </div>

</div>
```

```
<script src="js/app.js"></script>
</body>
</html>
```

Js/app.cs

```
// ---------- STATE ----------
let loggedIn = false;
let cart = [];

// ---------- PRODUCTS ----------
let products=[
  {id:1,             name:'Men             Shorts',             price:1199,
img:'https://images.pexels.com/photos/769749/pexels-photo-769749.jpeg'},
  {id:2,            name:'Women            Shirts',            price:1199,
img:'https://images.pexels.com/photos/3775087/pexels-photo-3775087.jpeg'},
  {id:3,            name:'Running            Shoes',            price:1200,
img:'https://images.pexels.com/photos/2529148/pexels-photo-2529148.jpeg'},
  {id:4,            name:'Sports            Shoes',            price:1500,
img:'https://images.pexels.com/photos/19090/pexels-photo.jpg'},
  {id:5,             name:'Casual             Shoes',             price:900,
img:'https://images.pexels.com/photos/2529147/pexels-photo-2529147.jpeg'},
  {id:6,              name:'Mobile              A',              price:12000,
img:'https://images.pexels.com/photos/607812/pexels-photo-607812.jpeg'},
  {id:7,              name:'Mobile              B',              price:15000,
img:'https://images.pexels.com/photos/1334597/pexels-photo-1334597.jpeg'},
  {id:8,              name:'Mobile              C',              price:18000,
img:'https://images.pexels.com/photos/699122/pexels-photo-699122.jpeg'},
  {id:9,              name:'Laptop              A',              price:42000,
img:'https://images.pexels.com/photos/18105/pexels-photo.jpg'},
  {id:10,             name:'Laptop              B',              price:55000,
img:'https://images.pexels.com/photos/374074/pexels-photo-374074.jpeg'}
];

// ---------- PAGE CONTROL ----------
function showPage(page){
  document.querySelectorAll('.page').forEach(p=>p.style.display='none');

  if((page=='catalog'||page=='cart') && !loggedIn){
    alert('Please login first');
    page='login';
  }
  document.getElementById(page).style.display='block';
}

// ---------- REGISTER ----------
function register(){
  regMsg.innerText = '';
  const username = regUser.value.trim();
  const password = regPass.value.trim();

  if(!username || !password){ regMsg.innerText='All fields are required';
return; }
```

```
  if(username.length<3){  regMsg.innerText='Username  must  be  at  least  3
characters'; return; }
  if(password.length<4){  regMsg.innerText='Password  must  be  at  least  4
characters'; return; }

  localStorage.setItem('user', username);
  localStorage.setItem('pass', password);
  regMsg.innerText='Registered successfully! Redirecting to login...';
  regMsg.style.color = 'green';

  regUser.value='';
  regPass.value='';

  // Redirect to login after 1 second
  setTimeout(() => {
    showPage('login');
    regMsg.innerText = '';
  }, 1000);


}

// ---------- LOGIN ----------
function login(){
  loginMsg.innerText='';
  const username=loginUser.value.trim();
  const password=loginPass.value.trim();
  if(!username){ loginMsg.innerText='Username is required'; return; }
  if(!password){ loginMsg.innerText='Password is required'; return; }

  const storedUser=localStorage.getItem('user');
  const storedPass=localStorage.getItem('pass');

  if(username===storedUser && password===storedPass){
    loggedIn=true;
    navLogin.classList.add('d-none');
    navRegister.classList.add('d-none');
    navLogout.classList.remove('d-none');
    loginUser.value=''; loginPass.value=''; loginMsg.innerText='';
    showPage('catalog');
  } else{
    loginMsg.innerText='Invalid credentials';
  }
}

// ---------- LOGOUT ----------
function logout(){
  loggedIn=false; cart=[];
  updateCart();
  navLogin.classList.remove('d-none');
  navRegister.classList.remove('d-none');
  navLogout.classList.add('d-none');
  showPage('login');
}
```

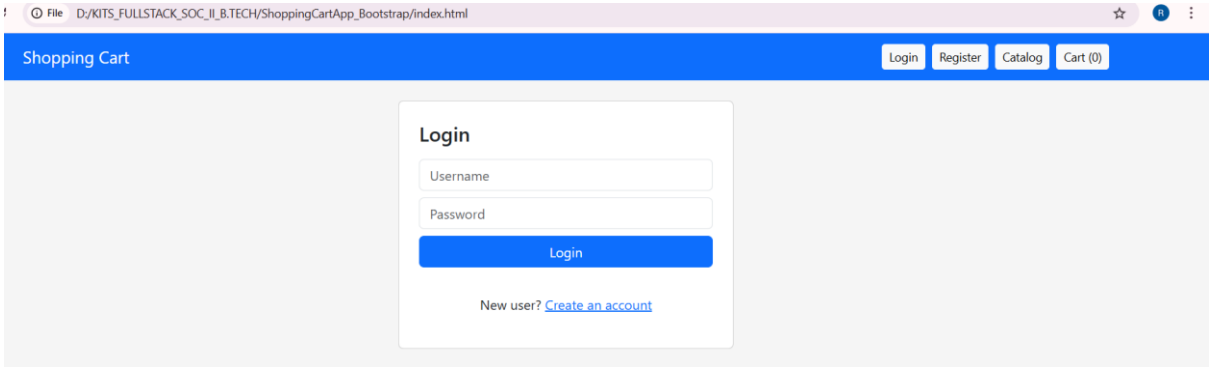
```
// ---------- LOAD PRODUCTS ----------
products.forEach(p=>{
  productList.innerHTML+=`
    <div class="col-12 col-sm-6 col-md-4 col-lg-3 mb-3">
      <div class="card text-center">
        <img src="${p.img}" class="card-img-top">
        <div class="card-body">
          <h6>${p.name}</h6>
          <p>₹${p.price}</p>
          <button          class="btn          btn-primary          btn-sm"
onclick="addToCart(${p.id})">Add to Cart</button>
        </div>
      </div>
    </div>
  `;
});

// ---------- CART ----------
function addToCart(id){
  let item=cart.find(i=>i.id===id);
  if(item) item.qty++;
  else{    let    p=products.find(p=>p.id===id);    cart.push({...p,qty:1});
alert(p.name+' added to cart'); }
  updateCart();
}

function updateCart(){
  cartItems.innerHTML=''; let total=0,count=0;
  cart.forEach((c,i)=>{
    total+=c.price*c.qty; count+=c.qty;
    cartItems.innerHTML+=`
      <div class="d-flex justify-content-between border-bottom py-2">
        <div>${c.name} - ₹${c.price} x ${c.qty}</div>
        <div>
          <button         class="btn         btn-sm         btn-secondary"
onclick="changeQty(${i},-1)">-</button>
          <button         class="btn         btn-sm         btn-secondary"
onclick="changeQty(${i},1)">+</button>
          <button          class="btn          btn-sm          btn-danger"
onclick="removeItem(${i})">Remove</button>
        </div>
      </div>`;
  });
  totalEl=document.getElementById('total');
countEl=document.getElementById('cartCount');
  totalEl.innerText=total; countEl.innerText=count;
}

function     changeQty(i,val){     cart[i].qty+=val;     if(cart[i].qty<=0)
cart.splice(i,1); updateCart(); }
function removeItem(i){ cart.splice(i,1); updateCart(); }

// ---------- START ----------
```

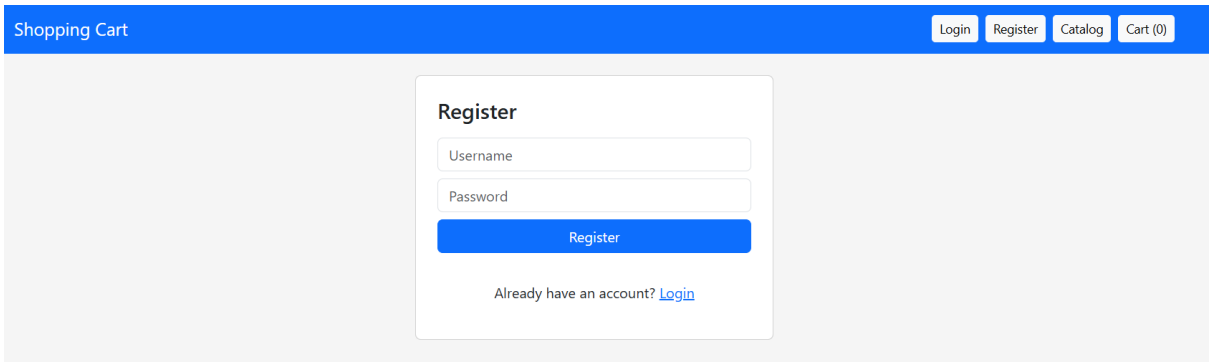
```
showPage('login');
```

<u>css/style.css</u>

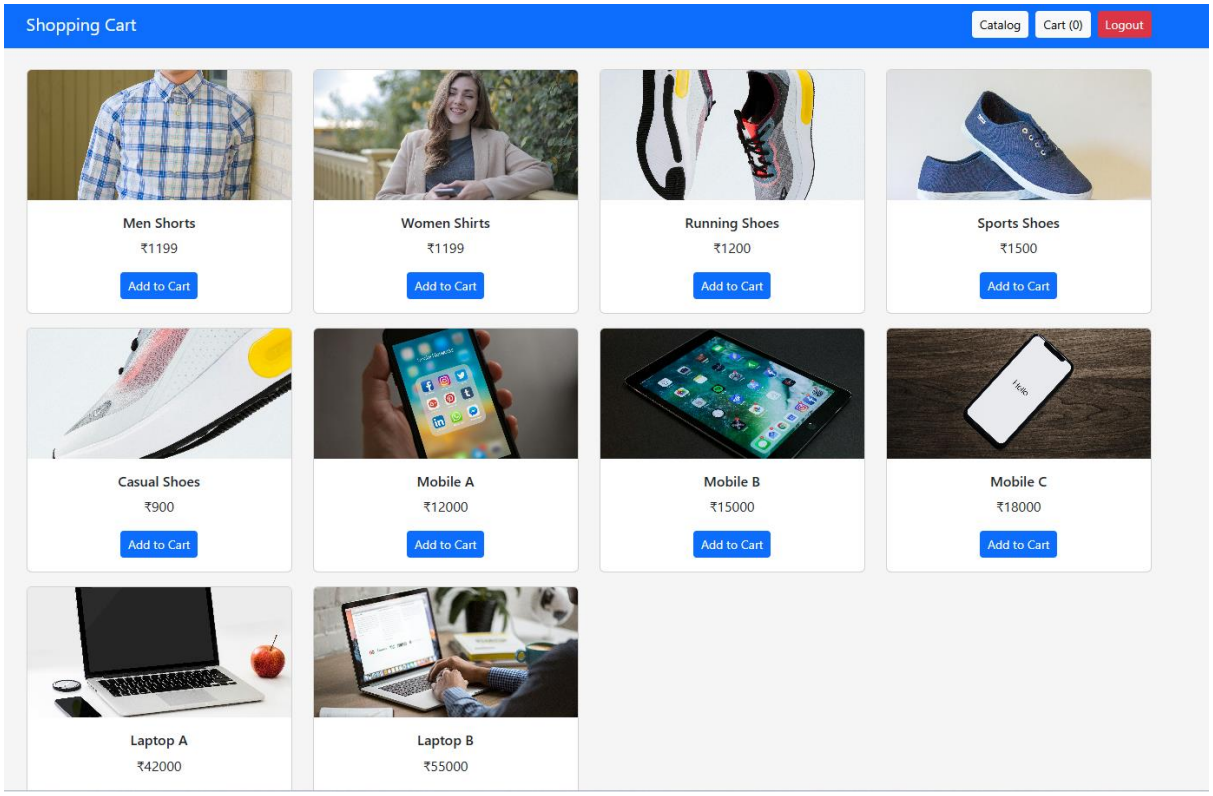
```
body { background:#f5f5f5 }
.page { display:none }
img { height:150px; object-fit:cover }
```

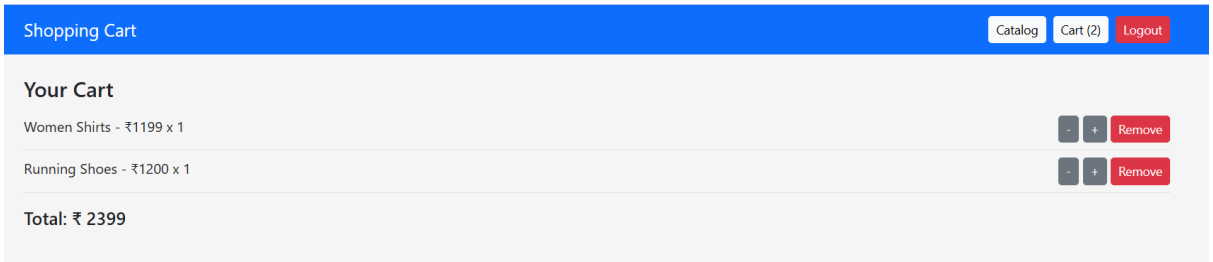
Output:



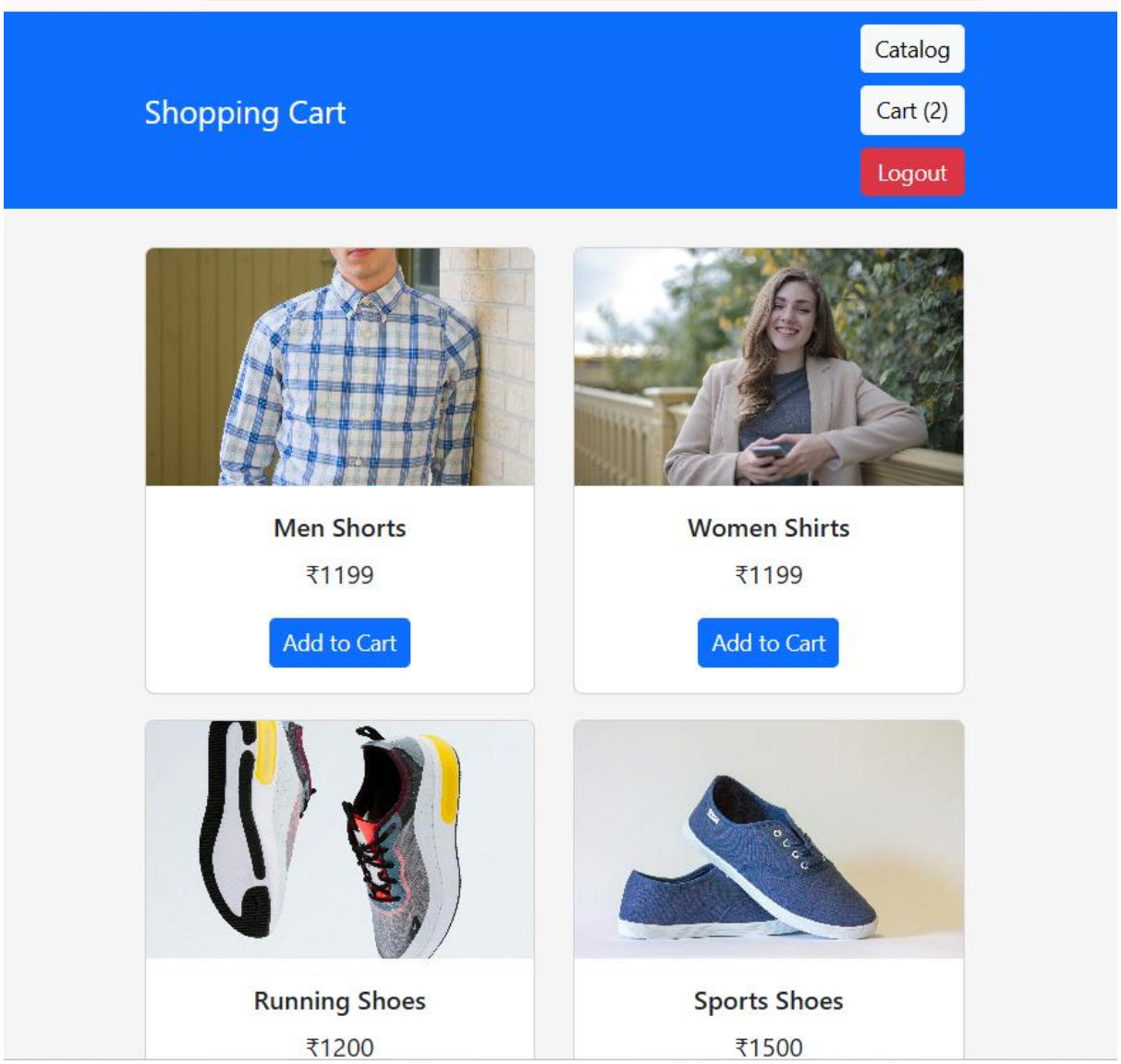


Laptop responsive web application





Tab/medium size responsive web application


Shopping Cart
Catalog
Cart (2)
Logout
Men Shorts
₹1199
Add to Cart
Women Shirts
₹1199
Add to Cart
Running Shoes
₹1200
Sports Shoes
₹1500

Mobile First responsive web application